Deriving CV = Ω/2 from the Two-Phase Structure of Coherence-Rupture-Regeneration Cycles

A Conjugate-Uncertainty Argument with Empirical Grounding

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Abstract. *The Coherence-Rupture-Regeneration (CRR) framework predicts coefficients of variation (CV) for oscillatory cycle times with no fitted parameters: CV = Ω/2, where Ω is determined by symmetry class (Ω_{Z2} = 1/π, $\Omega_{SO(2)}$ = 1/2π). Standard drift-diffusion first-passage theory gives CV = Ω, not Ω/2. Sabine (2026) identifies this as an open theoretical question, confirmed empirically across 132 systems at ~10.6σ significance. We propose a derivation grounded in the two-phase temporal structure of CRR: coherence (deterministic accumulation of the already-determined past) and regeneration (stochastic construction of the as-yet-undetermined future). We show that when these phases occupy equal mean time but contribute asymmetric variance, CV = Ω/2 follows. We contextualise this within recent results on time-information uncertainty relations (Nicholson et al., Nature Physics, 2020; Ito & Dechant, Phys. Rev. X, 2020) and propose testable next steps. An initial pre-registered prediction about distribution shape was tested and refuted, leading to the corrected derivation presented here.*

1. The CRR Framework: Canonical Equations

Coherence-Rupture-Regeneration (Sabine, 2024-2026) posits three operators governing the temporal dynamics of bounded systems. The framework has a single free parameter, Ω, which encodes the system's boundary permeability (capacity for flexible reorganisation).

Coherence — the past accumulates:

$$C(x,t) = \int_0^t L(x,\tau)\, d\tau$$

where L(x,τ) is the mnemonic entanglement rate. C is dimensionless, monotone non-decreasing within each cycle, and resets to 0 at each rupture.

Rupture — the present is instantaneous:

$$\delta(t - t^*) \quad \text{when} \quad C \cdot \Omega = 1$$

A Dirac delta marking the ontological present. Rupture occurs when accumulated coherence reaches threshold C* = 1/Ω. The rupture condition C·Ω = 1 is the Cramer-Rao bound of estimation theory: accumulated evidence times irreducible uncertainty equals unity.

Regeneration — the future is built from weighted memory:

$$R[\phi](x,t) = \int_0^t \phi(x,\tau) \cdot \exp(C(x,\tau)/\Omega) \cdot \Theta(t-\tau)\, d\tau \;\; / \;\; Z(t)$$

After rupture, the system reconstructs by weighting historical states by their coherence at each historical moment. The exponential kernel exp(C/Ω) ensures high-coherence moments contribute more, regardless of recency. At rupture, C = Ω, giving the Euler calibration: exp(Ω/Ω) = e ≈ 2.718.

Symmetry classes and predictions (no fitted parameters):

Symmetry	Phase to rupture	Ω	$CV = \Omega/2$	Example
Z_2 (bistable)	π radians	$1/\pi \approx 0.318$	$1/(2\pi) \approx 0.159$	Cardiac, neural, Ca^{2+}
SO(2) (rotational)	2π radians	$1/(2\pi) \approx 0.159$	$1/(4\pi) \approx 0.080$	Segmentation clock, gait

2. The Open Problem: Why CV = $\Omega/2$, Not Ω

For a drift-diffusion process $dC = \mu dt + \sigma dW$ hitting threshold $b = 1/\Omega$ with drift $\mu = 1$ and $\sigma^2 = \Omega$ (the FEP identification Ω = variance), the inverse Gaussian first-passage time gives:

$$CV^2 = \sigma^2 / (b \cdot \mu) = \Omega / (1/\Omega) = \Omega^2 \quad \Rightarrow \quad CV = \Omega$$

Empirically, CV = $\Omega/2$. This is confirmed across 132 systems at ~10.6σ significance (Sabine, 2026), but the factor of 1/2 lacks a rigorous derivation. Sabine notes it is 'consistent with the conjugacy structure that connects CRR to the Cramer-Rao bound, the Heisenberg uncertainty principle, and the Gabor limit' but calls the connecting proof 'not yet complete.'

3. Derivation: The Two-Phase Cycle

3.1 Key insight: coherence is deterministic, regeneration is stochastic

The CRR cycle consists of two temporal phases separated by the instantaneous rupture δ(now):

Coherence integrates the past: $C(x,t) = \int L(x,\tau)d\tau$. The past has already happened. It has content. The system reads from what is already determined, accumulating evidence that already exists. This is a guided, low-variance process. Coherence is the **deterministic** phase.

Regeneration constructs the future: $R = \int \phi \cdot \exp(C/\Omega) \cdot \Theta \, d\tau$. The $\exp(C/\Omega)$ kernel selects which historical patterns are accessible, but which specific configuration actually emerges from those weighted possibilities is not determined in advance. Multiple potential reconstructions compete. The system must settle on one. Regeneration is the **stochastic** phase.

This assignment aligns with Whiteheadian process philosophy (the past is determined, the future is genuinely open), with the Free Energy Principle (perception converges on data; action selects among possibilities), and with the observed positive skew of cycle-time distributions (the stochastic reconstruction phase, not the deterministic accumulation phase, generates the right tail).

3.2 The variance asymmetry

Let T_c be the coherence phase duration and T_r be the regeneration phase duration. The total cycle time is $T = T_c + T_r$.

$$\text{Coherence (deterministic):} \quad E[T_c] = 1/\Omega \qquad Var(T_c) \approx 0$$

$$\text{Regeneration (stochastic):} \quad E[T_r] = 1/\Omega \qquad CV(T_r) = \Omega$$

The assumption $E[T_c] = E[T_r]$ reflects the duality of the two phases: building structure and reconstructing structure are symmetric operations requiring comparable time. This is empirically supported by the approximate 50% duty cycles observed in cardiac (systole/diastole), neural (depolarisation/repolarisation), and respiratory (inspiration/expiration) oscillators.

3.3 The result

$$E[T] = E[T_c] + E[T_r] = 1/\Omega + 1/\Omega = 2/\Omega$$

$$Var(T) = Var(T_c) + Var(T_r) \approx 0 + Var(T_r)$$

$$SD(T_r) = CV(T_r) \times E[T_r] = \Omega \times 1/\Omega = 1$$

$$CV(T) = SD(T) / E[T] = 1 / (2/\Omega) = \mathbf{\Omega/2} \qquad \text{QED}$$

The factor of 2 in the denominator literally counts the two phases of the CRR cycle. At balance ($E[T_c] = E[T_r]$), exactly half the cycle is deterministic reading of the past, and half is stochastic construction of the future.

4. Physical Grounding: Why This Phase Assignment Is Correct

4.1 Fisher information decays during relaxation

Ito & Dechant (2020) prove in *Physical Review X* that for Markovian relaxation processes without explicit time dependence, the Fisher information is a monotonically decreasing function of time. Regeneration is a relaxation process: the system settling into a new configuration from weighted historical templates. As it relaxes, it becomes *less* informative, *less* variable, *more* deterministic. However, the initial phase of regeneration, when multiple competing configurations have comparable $\exp(C/\Omega)$ weights, is genuinely stochastic. The variability in regeneration time comes from the variable duration of this competition, not from noise during the settling phase.

4.2 Speed limits from the Cramer-Rao bound

Nicholson et al. (2020) prove in *Nature Physics* that rates of energy and entropy exchange obey a time-information uncertainty relation that mirrors the Mandelstam-Tamm bound in quantum mechanics. The timescale of transient fluctuations is universally bounded by the Fisher information. CRR's rupture condition $C \cdot \Omega = 1$ is a specific instance of this class of bounds, operating at saturation. The factor of 1/2 in $CV = \Omega/2$ is consistent with the minimum uncertainty product for conjugate variables at the information-theoretic boundary.

4.3 Biological oscillators suppress period variance below stochastic prediction

Potvin-Trottier et al. (2016, *PLOS Computational Biology*) show that in synthetic gene oscillators, intrinsic noise alone produces period CV five times higher than observed. Only by including extrinsic regulatory mechanisms can the model match data. This is consistent with the two-phase picture: the deterministic coherence phase absorbs cycle time without contributing proportional variance, suppressing overall CV below what pure stochastic accumulation would predict.

Ozbudak et al. (2023, *Nature Communications*) demonstrate that in the zebrafish segmentation clock, individual oscillation cycles stochastically fail to pass a threshold amplitude. The variability is in the commitment/reconstruction phase, not in the evidence accumulation phase.

5. Predictions for the Three-Class Taxonomy

The two-phase derivation refines CRR's three-class taxonomy by specifying the mechanism:

Class	Regeneration character	CV prediction	Mechanism
A (Autonomous)	Normal stochasticity $E[T_r] = E[T_c]$	$CV = \Omega/2$ (intrinsic value)	Balanced two-phase cycle
B (Regulated)	Constrained; few accessible configurations	$CV \ll \Omega/2$ (suppressed)	Regeneration fast + deterministic
C (Noisy/Volitional)	Extra stochastic; many competing options	$CV \gg \Omega/2$ (elevated)	Regeneration slow + high-variance

This is a more specific mechanistic claim than the original CRR benchmarks paper makes. It says that Class B systems have suppressed CV because regeneration is fast and constrained (few accessible patterns), while Class C systems have elevated CV because regeneration is slow and variable (many competing patterns). This is testable independently of CV itself, by examining the temporal structure *within* cycles.

6. Pre-Registered Prediction: Tested and Refuted

An earlier version of this derivation (based on conjugate uncertainty at the commitment event) predicted that cycle-time distributions should be approximately Gaussian (symmetric), with near-zero skewness. This prediction was pre-registered and tested against the literature:

The data is unambiguous. Reaction time distributions exhibit positive skew and are well-fit by ex-Gaussian and inverse Gaussian models (Luce, 1986; Van Zandt, 2000; Heathcote et al., 1991). Cardiac RR interval distributions are non-Gaussian with potential multimodality. The prediction of symmetric distributions was **refuted**.

This refutation was productive: it showed that the variability is generated during a stochastic process (consistent with a first-passage-time distribution), not by fixed additive noise at commitment. The corrected derivation (Section 3) preserves the numerical result CV = $\Omega/2$ while correctly predicting positively skewed cycle-time distributions, because the stochastic regeneration phase generates an inverse-Gaussian-like right tail.

7. What Remains Open

(a) The equal-time assumption. The derivation requires $E[T_c] = E[T_r]$. This is plausible for balanced oscillators (cardiac systole/diastole, neural depolarisation/repolarisation) but needs systematic verification across the 132 systems in the CRR benchmark. If the duty cycle deviates from 50%, the derivation generalises to $CV = \Omega \cdot f_r$, where f_r is the fraction of cycle time spent in regeneration. The prediction $CV = \Omega/2$ then becomes a prediction that $f_r = 1/2$ for autonomous oscillators.

(b) Formal identification of coherence and regeneration phases in data. The derivation requires distinguishing which portion of an empirical cycle corresponds to 'coherence' (reading the past) and which to 'regeneration' (constructing the future). In some systems this is straightforward (cardiac systole vs diastole, neural spike vs refractory period). In others it requires careful operational definition.

(c) The geometric origin of Ω. The Bonnet-Myers theorem on positively curved statistical manifolds gives $\Omega = \pi/\sqrt{\kappa}$. The value $\kappa = 1$ (unit curvature) is assumed but not derived from first principles. This is the same gap identified by Sabine (2026).

(d) Rigorous connection to stochastic thermodynamics. The time-information uncertainty relations of Ito & Dechant (2020) and Nicholson et al. (2020) provide the general framework. A formal proof that CRR's specific cycle structure saturates these bounds would close the derivation completely.

8. Proposed Next Steps

Step 1: Duty-cycle survey. For each of the 132 systems in the CRR benchmark, identify the 'accumulation' and 'reconstruction' phases and measure their mean durations. Test whether the ratio $E[T_c]/E[T_r]$ clusters around 1.0 for Class A systems, exceeds 1.0 for Class B (short regeneration), and falls below 1.0 for Class C (prolonged regeneration). This is the most direct test of the derivation.

Step 2: Within-cycle variance decomposition. For systems where single-cycle data is available (cardiac RR with ECG morphology, neural ISI with spike waveform, segmentation clock with reporter imaging), decompose cycle-time variance into contributions from each phase. The derivation predicts that nearly all variance resides in the regeneration phase.

Step 3: Generalised CV formula. If the duty cycle is not exactly 50%, derive the generalised formula $CV = \Omega \cdot f_r(1 + f_c/f_r \cdot \varepsilon)$, where ε captures residual variance in the coherence phase. Test whether CV/Ω correlates with measured f_r across systems.

Step 4: Connection to Fisher information decay. Using the Ito-Dechant framework, compute the Fisher information trajectory through one full CRR cycle. The derivation predicts high Fisher information during coherence (deterministic, informative) and monotonically decreasing Fisher information during regeneration (relaxation). This can be tested in systems with continuous state measurements.

Step 5: Disrupted regeneration. Identify systems where regeneration is known to be impaired (e.g., pathological memory access, damaged feedback loops, early-cycle systems with little history). The derivation predicts these should show CV closer to Ω than to Ω/2, because the regeneration phase loses its deterministic character when historical access is limited.

9. References

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